

• Inner product space over field of real numbers.

Inner product defined over complex vector space.

Let V be a vector space over $\mathbb{C}$. A map $\beta: V \times V \rightarrow \mathbb{C}$ is called inner product if it

satisfies (i) $\beta(v_1 + v_2, w) = \beta(v_1, w) + \beta(v_2, w)$ } linear in first co-ordinate

(ii) $\beta(\alpha v, w) = \alpha \beta(v, w)$

(iii) $\beta(v, w_1 + w_2) = \beta(v, w_1) + \beta(v, w_2)$ } conjugate linear in second co-ordinate

(iv) $\beta(v, \alpha w) = \bar{\alpha} \beta(v, w)$

(v) $\beta(v, w) = \overline{\beta(w, v)}$ → conjugate symmetry

positive definite

(vi)

$\beta(v, v) \in \mathbb{R}$

and $\beta(v, v) \geq 0 \forall v \in V$,
& $\beta(v, v) = 0$ iff $v = 0$

Why so

$\mathbb{R}^2$
dot product

$(x_1, x_2) \cdot (y_1, y_2) = x_1 y_1 + x_2 y_2$

correct inner product $\mathbb{C}^2$

$(x_1, x_2) \cdot (y_1, y_2) = x_1 \bar{y}_1 + x_2 \bar{y}_2$

~~$\mathbb{C}^2$
inner product~~

~~$(x_1, x_2) \cdot (y_1, y_2) = x_1 y_1 + x_2 y_2$~~

~~$\textcircled{\mathbb{C}}, \quad \begin{matrix} i \cdot i = -1 \\ |i| = \sqrt{-1} \end{matrix} \rightarrow \text{Not correct.}$~~

$\textcircled{\mathbb{C}} \quad |i| = \sqrt{\langle i, i \rangle} = \sqrt{i(-i)} = \sqrt{1} = 1$

Example:- Define $\mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ by

$\langle (x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \rangle = x_1 \bar{y}_1 + x_2 \bar{y}_2 + \dots + x_n \bar{y}_n$

show that the above map is an inner product on $\mathbb{C}^n$.

ex 2 ① $\langle (x_1, x_2) + (x'_1, x'_2), (y_1, y_2) \rangle = \langle (x_1, x_2), (y_1, y_2) \rangle + \langle (x'_1, x'_2), (y_1, y_2) \rangle.$

② $\langle \alpha(x_1, x_2), (y_1, y_2) \rangle = \alpha \langle (x_1, x_2), (y_1, y_2) \rangle.$

③ $\langle (x_1, x_2), (y_1, y_2) + (y'_1, y'_2) \rangle = \langle (x_1, x_2), (y_1, y_2) \rangle + \langle (x_1, x_2), (y'_1, y'_2) \rangle.$

④ $\begin{aligned} \langle (x_1, x_2), \alpha(y_1, y_2) \rangle &= \langle (x_1, x_2), (\alpha y_1, \alpha y_2) \rangle \\ &= x_1(\alpha \bar{y}_1) + x_2(\alpha \bar{y}_2) \\ &= x_1 \alpha \bar{y}_1 + x_2 \alpha \bar{y}_2 \\ &= \alpha (x_1 \bar{y}_1 + x_2 \bar{y}_2) = \alpha \langle (x_1, x_2), (y_1, y_2) \rangle \end{aligned}$

⑤ check $\langle (x_1, x_2), \alpha(y_1, y_2) \rangle = \overline{\langle (y_1, y_2), (x_1, x_2) \rangle}$

$$\begin{aligned} \langle (x_1, x_2), (y_1, y_2) \rangle &= x_1 \bar{y}_1 + x_2 \bar{y}_2 \\ &= \overline{y_1 \bar{x}_1 + y_2 \bar{x}_2} = \overline{y_1 \bar{x}_1 + y_2 \bar{x}_2} \\ &= \langle (y_1, y_2), (x_1, x_2) \rangle \end{aligned}$$

⑥ positive definite :- (i) $\langle (x_1, x_2), (x_1, x_2) \rangle \in \mathbb{R}$

(ii) $\langle (x_1, x_2), (x_1, x_2) \rangle \geq 0$

(iii) $\langle (x_1, x_2), (x_1, x_2) \rangle = 0 \iff (x_1, x_2) = 0$

$x_1 \in \mathbb{C}, \quad x_2 \in \mathbb{C}$
 $x_1 = \alpha_1 + i\beta_1, \quad x_2 = \alpha_2 + i\beta_2, \quad \alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$

$$\begin{aligned} \langle (x_1, x_2), (x_1, x_2) \rangle &= x_1 \bar{x}_1 + x_2 \bar{x}_2 \\ &= (\alpha_1 + i\beta_1)(\alpha_1 - i\beta_1) + (\alpha_2 + i\beta_2)(\alpha_2 - i\beta_2) \end{aligned}$$

$$\begin{aligned}
 \langle (x_1, x_2), (x_1, x_2) \rangle &= x_1 x_1 + x_2 x_2 \\
 &= (\alpha_1 + i\beta_1)(\alpha_1 - i\beta_1) + (\alpha_2 + i\beta_2)(\alpha_2 - i\beta_2) \\
 &= \alpha_1^2 + \beta_1^2 + \alpha_2^2 + \beta_2^2 \geq 0
 \end{aligned}$$

$$\langle (x_1, x_2), (x_1, x_2) \rangle = 0 \quad \text{iff} \quad \begin{cases} \alpha_1 = 0, \alpha_2 = 0, \beta_1 = 0, \beta_2 = 0 \\ (x_1, x_2) = (0, 0) \end{cases}$$

• $\mathbb{C}^2$ is an inner product space.

Orthonormal set :- Let V be an inner product space over F . A subset $S = \{v_1, v_2, \dots, v_k\}$ is called

Orthonormal set if (i) $\|v_i\| = 1 \quad \forall \quad 1 \leq i \leq k$

(ii) $\langle v_i, v_j \rangle = 0 \quad \forall \quad 1 \leq i, j \leq k, \quad i \neq j.$

* Any orthonormal set is a linearly independent set.

* Maximal linearly independent set is a basis.

Orthonormal Basis :- Let V be an inner product space over F . A orthonormal set, which is also a basis of V , is called an orthonormal basis.

Example :- $\mathbb{R}^2$, Dot product is an inner product space

$\{(1, 0), (0, 1)\}$ is an orthonormal basis

$$|(1, 0)| = \sqrt{1^2 + 0^2} = \sqrt{1} = 1$$

$$|(0, 1)| = \sqrt{0^2 + 1^2} = \sqrt{1} = 1$$

$$\langle (1,0), (0,1) \rangle = 1 \cdot 0 + 0 \cdot 1 = 0$$

- $\left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \right\}$ is also an orthonormal Basis.

question:- Find the orthonormal Basis of subspace of $\mathbb{R}^3$ spanned by set $\{(1,1,0), (2,1,1)\}$.

- How to find orthonormal Basis of subspace of V spanned by set $\{v_1, v_2, \dots, v_k\}$.

- e_1 $\left\{ \frac{v_1}{\|v_1\|} \right\}$ (i) Orthonormal Basis for subspace spanned by $\{v_1\}$ subspace spanned by $\{v_2, e_1\}$.
- (ii) Orthonormal Basis for (subspace spanned by $\{v_1, v_2\}$)
- $\vdots$
- (k)th Orthonormal Basis for subspace spanned by $\{v_1, v_2, \dots, v_k\}$.

* Orthonormal Basis for the subspace spanned by $\{(1,1,0)\}$ is $\left\{ \frac{(1,1,0)}{\|(1,1,0)\|} \right\}$

$$\|(1,1,0)\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2} \quad e_1$$

In this case, orthonormal Basis is $\left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \right\}$.

* Orthonormal Basis for the subspace spanned by $\{v_2, e_1\}$

we need to find orthonormal Basis $\{e_2, e_1\}$ ✓

we need to find orthonormal basis $\{e_2, e_1\}$
 s.t. (i) $e_2 \in \text{L.S. } \{v_2, e_1\}$

(ii) $\|e_2\| = 1$ (iii) $\langle e_1, e_2 \rangle = 0$

Suppose :

$$w_2 = v_2 - \langle v_2, e_1 \rangle e_1 \in \text{L.S. } \{v_2, e_1\}$$

$$\begin{aligned} \langle w_2, e_1 \rangle &= \langle v_2 - \langle v_2, e_1 \rangle e_1, e_1 \rangle \\ &= \langle v_2, e_1 \rangle - \langle v_2, e_1 \rangle \underbrace{\langle e_1, e_1 \rangle}_{=1} \\ &= \langle v_2, e_1 \rangle - \langle v_2, e_1 \rangle = 0 \end{aligned}$$

$$e_2 = \frac{w_2}{\|w_2\|}, \quad \langle e_2, e_1 \rangle = \left\langle \frac{w_2}{\|w_2\|}, e_1 \right\rangle = \frac{1}{\|w_2\|} \langle w_2, e_1 \rangle = 0.$$

• Orthonormal Basis for subspace spanned by

$$\begin{aligned} &\{ (1, 1, 0), (2, 1, 1) \} \\ &= \text{subspace spanned by } \left\{ \underbrace{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)}_{e_1}, \underbrace{(2, 1, 1)}_{v_2} \right\} \end{aligned}$$

$$w_2 = v_2 - \langle v_2, e_1 \rangle e_1$$

$$w_2 = (2, 1, 1) - \left\langle (2, 1, 1), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \right\rangle \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$w_2 = (2, 1, 1) - \frac{3}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) = (2, 1, 1) - \left(\frac{3}{2}, \frac{3}{2}, 0 \right)$$

$$w_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1 \right)$$

$$\|w_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$$

$$e_2 = \frac{w_2}{\|w_2\|} = \sqrt{\frac{2}{3}} \left(\frac{1}{2}, -\frac{1}{2}, 1 \right) = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}} \right)$$

$$\begin{aligned} \langle e_1, e_2 \rangle &= \left\langle \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right), \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}} \right) \right\rangle \\ &= \frac{1}{\sqrt{2} \times \sqrt{6}} - \frac{1}{\sqrt{2} \times \sqrt{6}} = 0. \end{aligned}$$

Algorithm Basis	orthonormal Basis
$\{v_1\}$	$\left\{ \frac{v_1}{\ v_1\ } \right\}, e_1 = \frac{v_1}{\ v_1\ }, \{e_1\}.$
$\{v_1, v_2\}$	$w_2 = v_2 - \langle v_2, e_1 \rangle e_1$ $e_2 = \frac{w_2}{\ w_2\ }, \{e_1, e_2\}$
$\{v_1, v_2, v_3\}$	$\{e_1, e_2, e_3\}$ $w_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$ $e_3 = \frac{w_3}{\ w_3\ }$

* Subspace spanned by $\{v_1, v_2, v_3\}$ is same as subspace spanned by $\{e_1, e_2, v_3\}$

To find $e_3 \in \text{L.S } \{e_1, e_2, v_3\}$

$$\|e_3\| = 1$$

$$\langle e_1, e_3 \rangle = 0, \quad \langle e_2, e_3 \rangle = 0$$

$$\boxed{\langle e_1, e_3 \rangle = 0, \quad \langle e_2, e_3 \rangle = 0}$$

$$w_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2$$

$$\langle w_3, e_1 \rangle = \langle v_3, e_1 \rangle - \langle v_3, e_1 \rangle \underbrace{\langle e_1, e_1 \rangle}_{=1} - \langle v_3, e_2 \rangle \underbrace{\langle e_2, e_1 \rangle}_{=0}$$

$$= \langle v_3, e_1 \rangle - \langle v_3, e_1 \rangle = 0$$

$$\langle w_3, e_2 \rangle = \langle v_3, e_2 \rangle - \langle v_3, e_1 \rangle \underbrace{\langle e_1, e_2 \rangle}_{=0} - \langle v_3, e_2 \rangle \underbrace{\langle e_2, e_2 \rangle}_{=1}$$

$$\langle w_3, e_2 \rangle = 0$$

$$, \quad \boxed{w_3 / \|w_3\| = e_3}$$

similarly

$e_1, e_2, \dots, e_{k-1},$

$\boxed{v_k}$

$$w_k = v_k - \sum_{i=1}^{k-1} \langle v_k, e_i \rangle e_i$$

$$e_k = \frac{w_k}{\|w_k\|}$$

Gram
Schmidt
orthogonalisation